

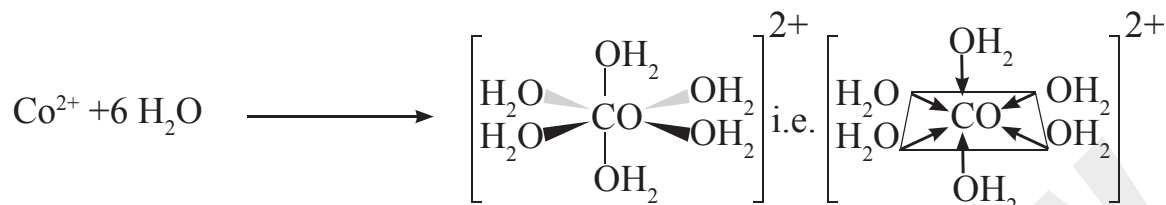
Experiment No. 14

Aim: To study the shift in equilibrium between hexaaquacobalt (II) ion $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and chloride ions (Cl^-) by changing the concentration of either of the ions.

Theory: When CoCl_2 is dissolved in water it ionizes as follows.



Ionized Co^{2+} ions form a complex $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ ion with water.



The complex in presence of Cl^- ions forms $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_2$ and the color of this complex is pink. The intensity of the color depends on the concentration of the complex. The equation can be displayed by changing the concentrations of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and Cl^- ions. In general increase in concentration of reactants shifts the equilibrium in the direction of products and vice versa.

Apparatus: Burette Five 100 mL beakers, test tube, test tube stand

Chemicals: CoCl_2 (0.1M), HCl (0.1M) (solutions should be provided to students), distilled water

Reaction:



Procedure:

1. Mix CoCl_2 (0.1M), HCl (0.1M) and distilled water in five different 100 ml beakers in volumes as shown in the following table.

Beaker	0.1M CoCl_2 mL	0.1M HCl mL	Distilled water mL	Intensity of color
I	10.0	20.0	10.0	Lighter
II	10.0	30.0	0.0	Deeper light
III	10.0	10.0	20.0	Blue
IV	15.0	20.0	5.0	Deeper light
V	5.0	20.0	15.0	lighter

2. Mix well solutions of every beaker and transfer it to five different test tubes.

3. Compare the colors developed in five different test tubes.

Conclusion: The balanced chemical equation suggests that for obtaining better yield of the product $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_2$ (pink color) reactant concentration of CoCl_2 and HCl in the ratio 1:2.

Therefore the intensity of color in beaker I is the ideal. The intensity of color in beaker II and IV will increase indicating in concentration of product. Whereas in beaker III and V will decrease indicating decrease in concentration of product.

Remark and sign of teacher:

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. The total volume of 0.1M CoCl_2 , 0.1M HCl and distilled water in each beaker is
a. 10 mL ✓ b. 20 mL c. 30 mL d. 40 mL
2. The color of solution of mixture of 0.1M CoCl_2 and 0.1M HCl is
a. Red ✓ b. Pink c. Blue d. Yellow
3. What is the ratio of reactant concentration of CoCl_2 and HCl for obtaining better yield?
a. 1:1 ✓ b. 1:2 c. 2:1 d. 1:3
4. In which one of the beaker the intensity of color is an ideal?
a. I b. II ✓ c. III d. IV

Short answer questions

1. What is chemical equilibrium?

Ans. Chemical equilibrium is the state of a reversible reaction in which the rate of the forward reaction becomes equal to the rate of the backward reaction and the concentrations of reactants and products remain constant.

2. Which color obtained when $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ treated with HCl?

Ans. A blue-colored complex is obtained due to the formation of chlorocobaltate complex ions.

3. What is ratio of concentration of (0.1M) HCl and (0.1 M) CoCl_2 is taken for better yield of $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_2$?

Ans. The ratio of HCl : CoCl_2 is 2 : 1.

4. What is effect of catalyst on chemical equilibrium?

Ans. A catalyst does not change the position of equilibrium. It only speeds up both the forward and backward reactions equally, helping the system reach equilibrium faster.

5. Chemical equilibrium is dynamic in nature. Justify your answer.

Ans. At equilibrium, the forward and backward reactions continue simultaneously at equal rates. Therefore, no observable change occurs, although the reactions are still taking place.

6. Does law of mass action apply on the following reaction



Ans. Yes, the law of mass action applies to this reversible reaction. The equilibrium depends on the concentrations of the reactants and products.

7. How does the concentration of reactant affect direction of chemical equilibrium?

Ans. According to Le Chatelier's principle, increasing the concentration of a reactant shifts the equilibrium in the forward direction, while decreasing it shifts the equilibrium in the backward direction.

Remark and sign of teacher: